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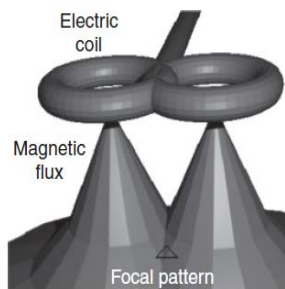
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Chapter 3

Recording and Stimulating the Brain

Minghui Shi

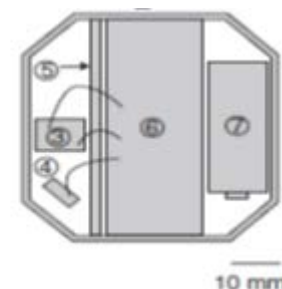
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Processing Stages --- Review

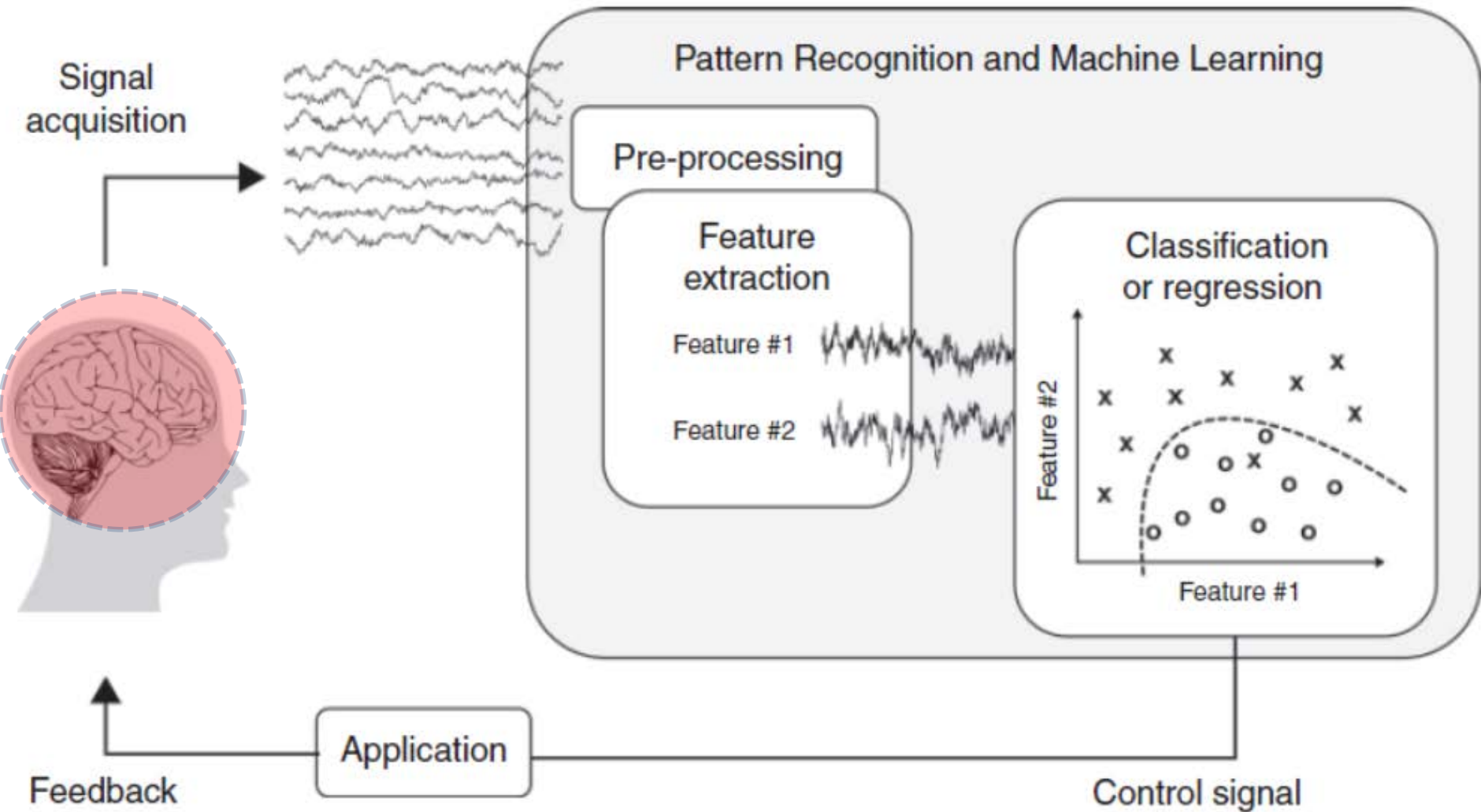


Figure 1.1. **Basic components of a brain-computer interface (BCI).**

Processing Stages --- Review

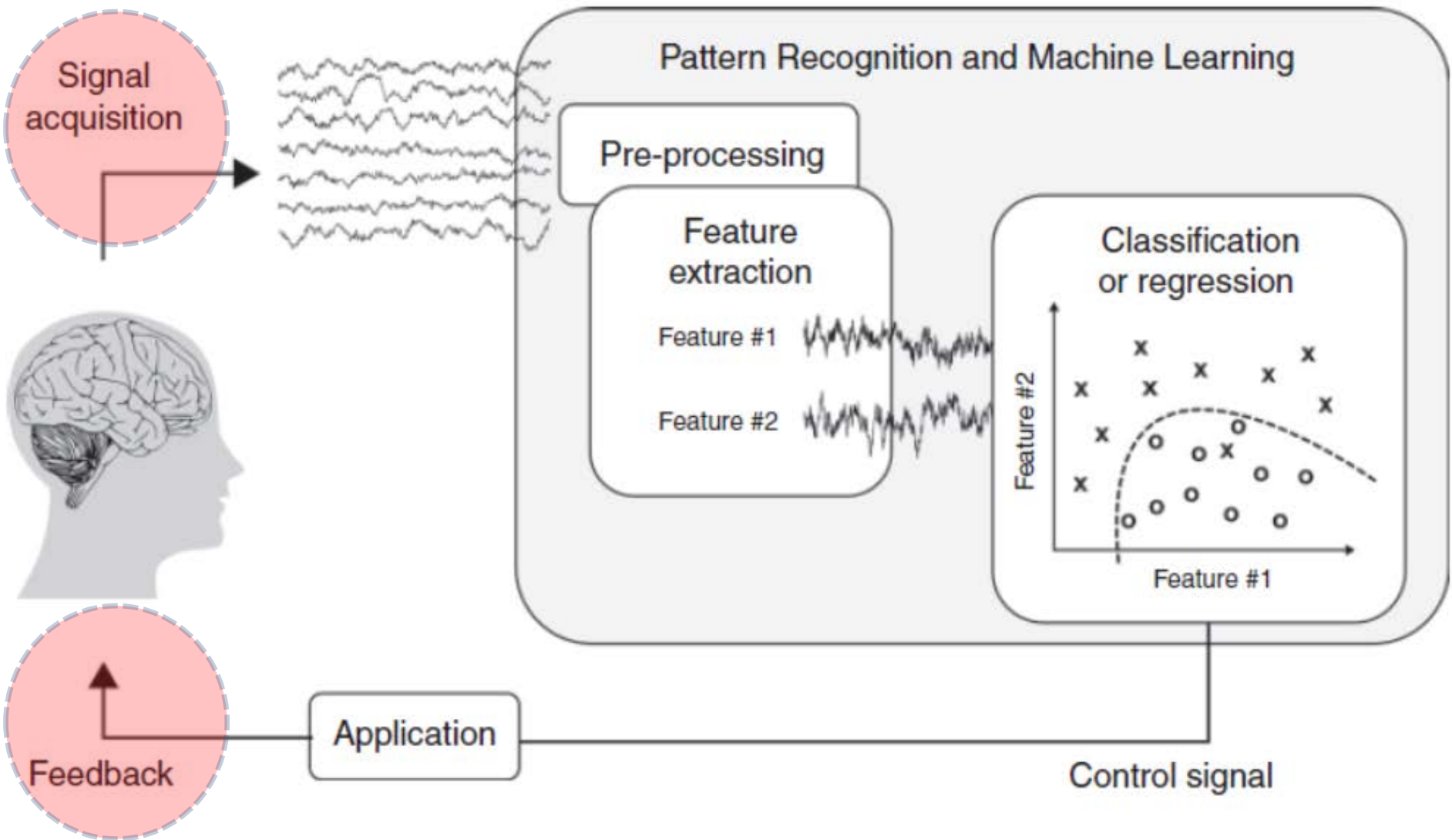


Figure 1.1. Basic components of a brain-computer interface (BCI).

Content & Topics

- **1. Introduction**
- **Part I Background**
 - 2. Basic Neuroscience
 - 3. Recording and Stimulating the Brain
 - 4. Signal Processing
 - 5. Machine Learning
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Goal

- Review some technologies that
 - **record** brain signals for BCIs.
 - **stimulate** neurons or brain regions.

Content

- 3.1 Recording Signals from the Brain
 - 3.1.1 Invasive Techniques
 - (Intracellular recording; Extracellular recording)
 - 3.1.2 Noninvasive Techniques
 - (EEG; MEG; fMRI; fNIR; PET)
- 3.2 Stimulating the Brain
 - 3.2.1 Invasive Techniques
 - (Microelectrodes; Direct Cortical Electrical Stimulation DCES; Optical Stimulation)
 - 3.2.2 Noninvasive Techniques
 - (Transcranial Magnetic Stimulation TMS; Transcranial Ultrasound)
- 3.3 Simultaneous Recording and Stimulation
 - 3.3.1 Multielectrode Arrays
 - 3.3.2 Neurochip

3.1.1 Invasive Techniques

3.1.1 Invasive Techniques

- **Video**

- **Invasive BCI**

- [Control devices and check emails with invasive BCI](#)
 - [Monkey's robotic arm](#)
 - [Invasive BCI research from Stanford University](#)
 - [Elon Musk hopes to put a computer chip in your brain.](#)

3.1.1 Invasive Techniques

1. What does the invasive recording method mean?
2. Can invasive techniques be used for human beings?
3. What are the differences when invasive techniques are used to human beings and animals?
4. Can invasive techniques cause pain?
5. Can invasive techniques be used for awaken animals?
6. What is the main advantage of invasive technologies, compared to non-invasive technologies?
7. Which tool is used to the invasive recording?

What does the invasive recording method mean?

- The invasive recording method can be used to record signals from **individual neurons** in the brain, and involves some form of **surgery**, wherein a part of the skull is removed, an electrode or implant **placed in the brain**, and the removed part of the skull then replaced.

Can invasive techniques be used for human beings?

- Yes, invasive techniques can be used for human beings.

What are the differences when invasive techniques are used to human beings and animals?

1. In the case of humans, invasive recordings are taken only in clinical settings such as during brain surgery or when patients are being monitored for abnormal brain activity (e.g., seizures) prior to surgery.
2. the time period available for recording may range from weeks and months to years in the case of some animals (e.g., monkeys) to a few days or minutes in the case of humans in a clinical setting.

Can invasive techniques cause pain?

- The recording itself is not painful, because the brain has no internal pain receptors.
- The surgery and recovery process can cause pain and involves risks such as infection.

Can invasive techniques be used for awoken animals?

- Yes. The recording procedure can be performed on both anesthetized as well as awake animals, although in the case of awoken recordings, the animal is typically restrained to minimize artifacts resulting from large movements.

What is the major advantage of invasive technologies, compared to non-invasive technologies?

- **High time resolution.**

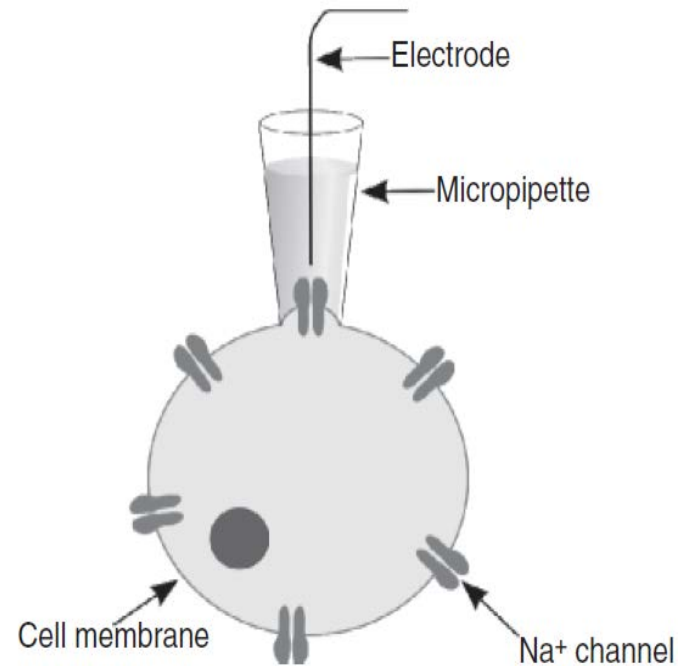
- A major advantage of invasive recordings is that they allow recording of action potentials at the millisecond timescale.
- This contrasts with noninvasive techniques, which measure indirect correlates of neural activity, such as blood flow, that occur at a coarser timescale (hundreds of milliseconds).

Which kind of tool is used to the invasive recording?

- **Microelectrode**, which is simply a very fine wire or other electrical conductor used to make contact with brain tissue.

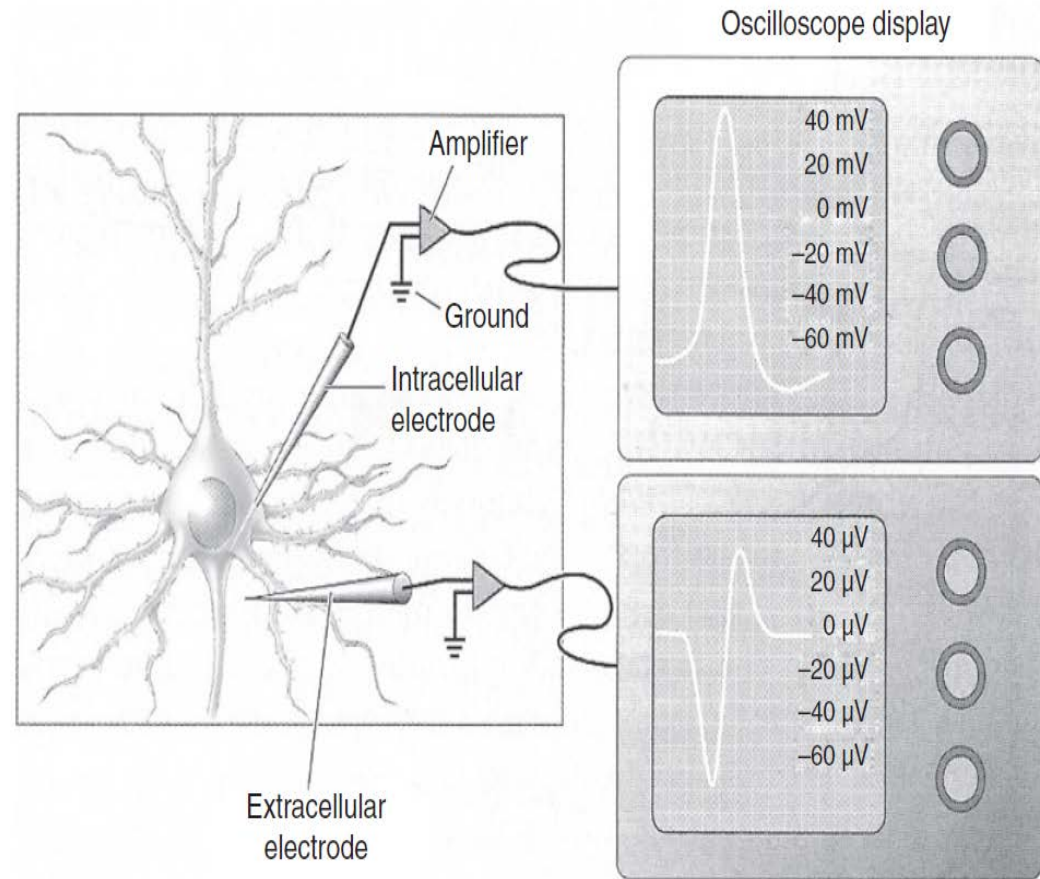
Intracellular recording

- **Figure 3.1. Intracellular recording using the patch clamp technique.**
- The technique allows measurement of ionic currents in a small patch of a cell membrane or the entire cell.



Intracellular versus extracellular recording

- **Figure 3.2. Intracellular versus extracellular recording of spikes.**
- The two oscilloscope displays on the right compare action potentials (spikes) recorded using intracellular (top) and extracellular (bottom) recording.



Intracellular V.S. extracellular recording (Con.)

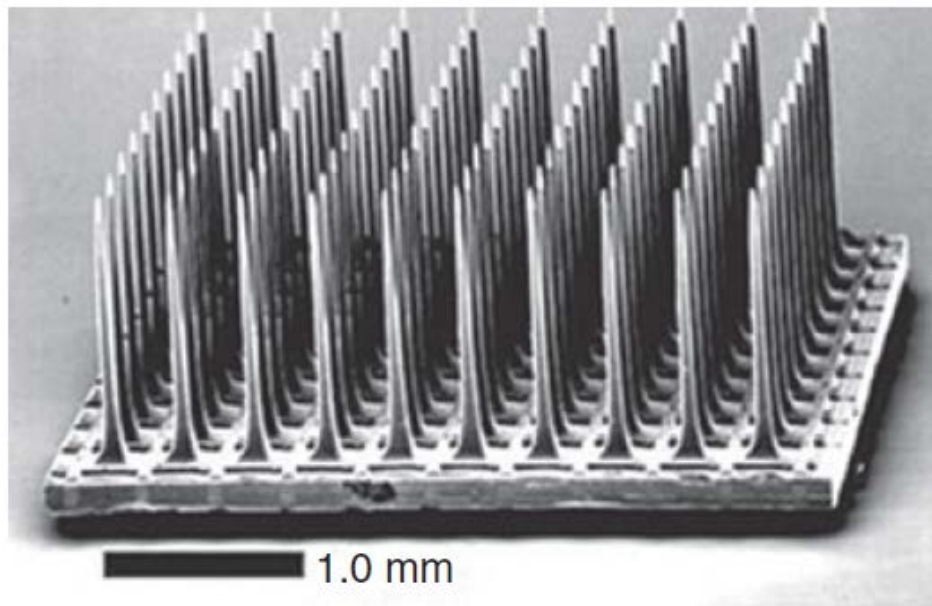
- **Intracellular recording** measures the potential difference between the inside of the cell (tip of the intracellular electrode) and an external electrode placed in the solution bathing the neuron (“ground”).
- **Extracellular recording** measures the potential difference between the tip of the extracellular electrode (placed near but outside a neuron) and a ground electrode.
- When the neuron produces a spike, positive ions flow away from the extracellular electrode into the neuron, causing the initial negative deflection in the display.
- This is followed by a positive deflection as the action potential decreases and positive charges flow out of the neuron toward the extracellular electrode.

Multielectrode Array

Multielectrode Array

Video:

1. [Control devices and check emails with invasive BCI](#)
2. [Invasive BCI research from Stanford University](#)



- Figure 3.3. **Example of a multielectrode array.**

What's the **advantage** of *multielectrode* arrays over more conventional *single-electrode* systems?

1. Increased spatial resolution;
2. Extracting complex types of information.

What's the **disadvantage** of *multielectrode* arrays over more conventional *single-electrode* systems?

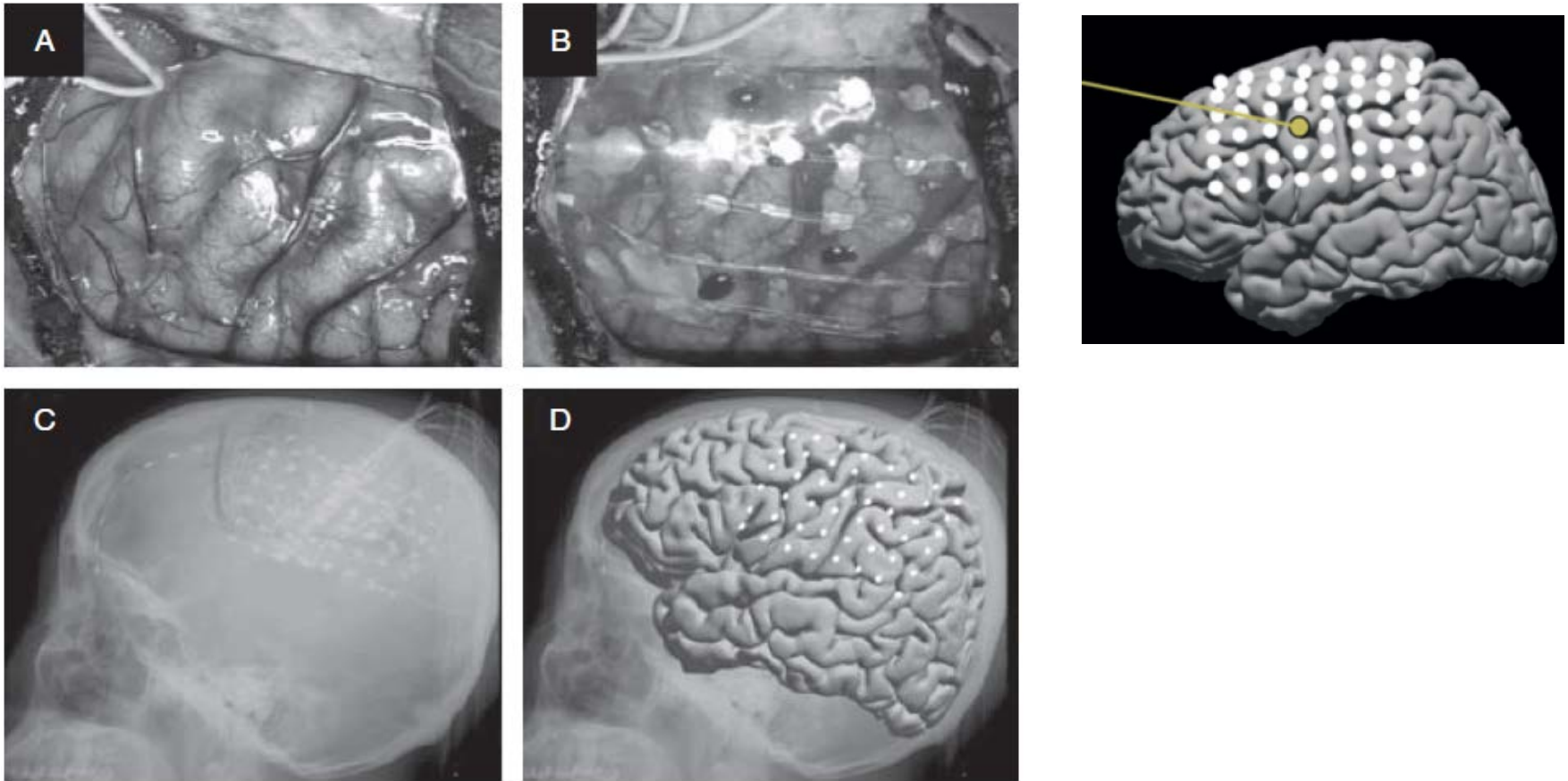
1. Increasing impedance of the electrodes.
 2. Significant reduction in signal quality over time.
- **Solution:** Ongoing research on biocompatibility of implants seeks to address these problems by coating the devices with polymers and other substances.

Electrocorticography (ECoG)

What is ECoG?

- **Electrocorticography (ECoG)** is a technique for recording brain signals that involves placing electrodes on the surface of the brain.
- The procedure requires making a surgical incision into the skull to implant the electrodes on the brain surface (Figure 3.4).

What is ECoG? (Con.)



- **Figure 3.4. ECoG in a human.**

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MicroECoG

- **Description:**

- MicroECoG electrodes are only a fraction of a millimeter in diameter and spaced only 2-3 mm apart in a grid.

- **Advantage:**

1. Much finer resolution
2. Decoding fine movements

- [ECoG video from Schalk Lab](#)

3.1.2 Noninvasive Techniques

EEG / MEG / FMRI / FNIR / PET

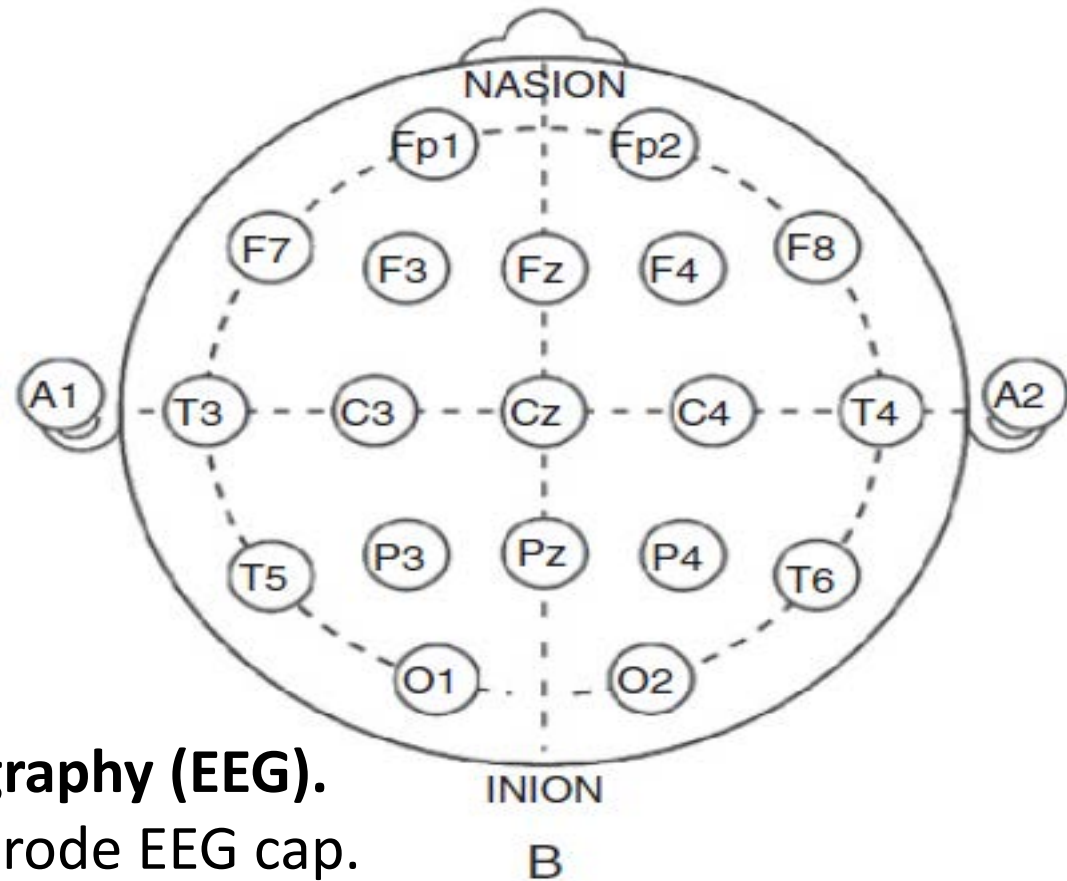
Electroencephalography (EEG)

What is EEG?

- **Electroencephalography (EEG)** is a popular noninvasive technique for recording signals from the brain using electrodes placed on the scalp.



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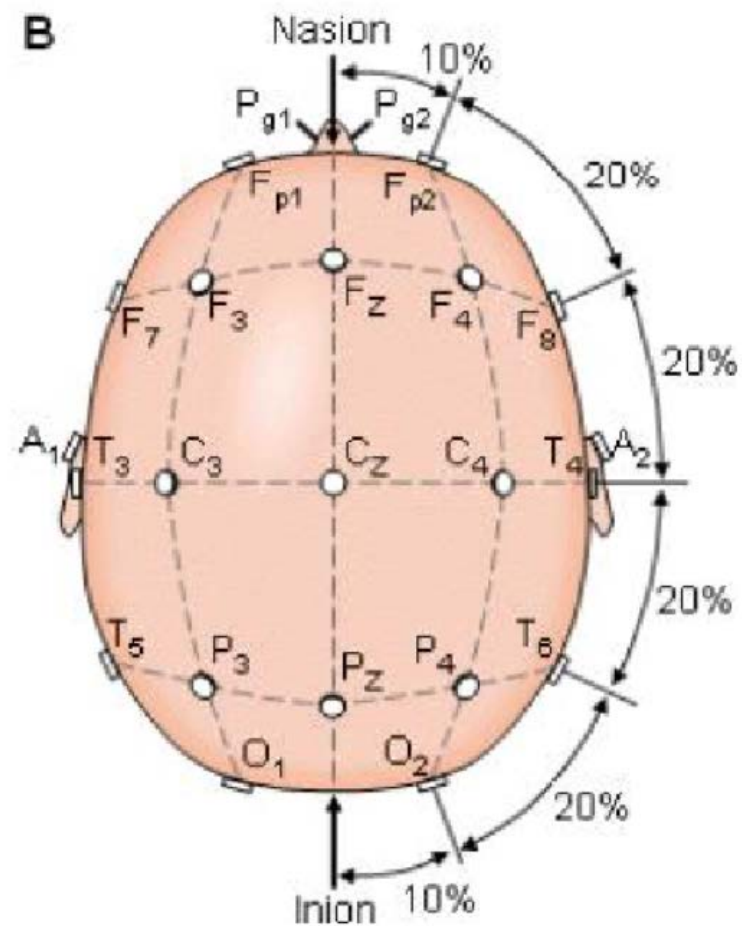
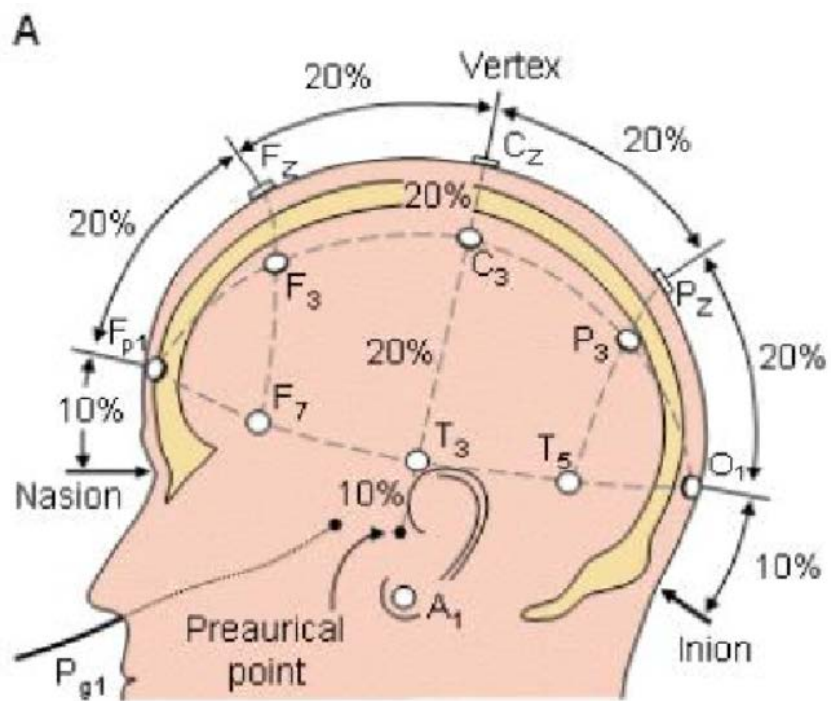
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Figure 3.7. **Electroencephalography (EEG).**

(A) Subject wearing a 32- electrode EEG cap.

(B) International 10–20 system for standardized EEG electrode locations on the head.

C = central, P = parietal, T = temporal, F = frontal, Fp = frontal polar, O = occipital, A = mastoids (image A courtesy K. Miller; image B from Wikimedia Commons).





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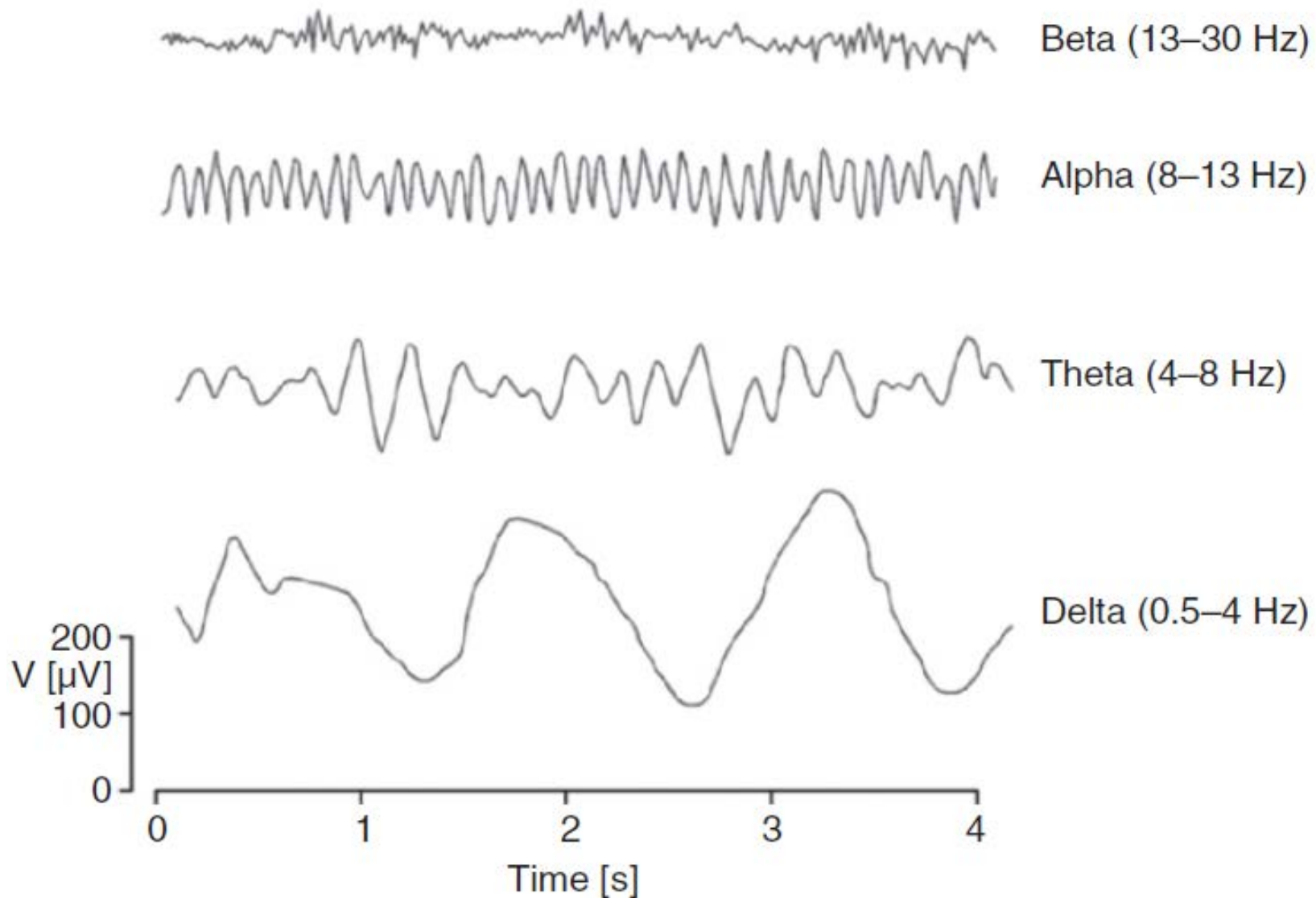


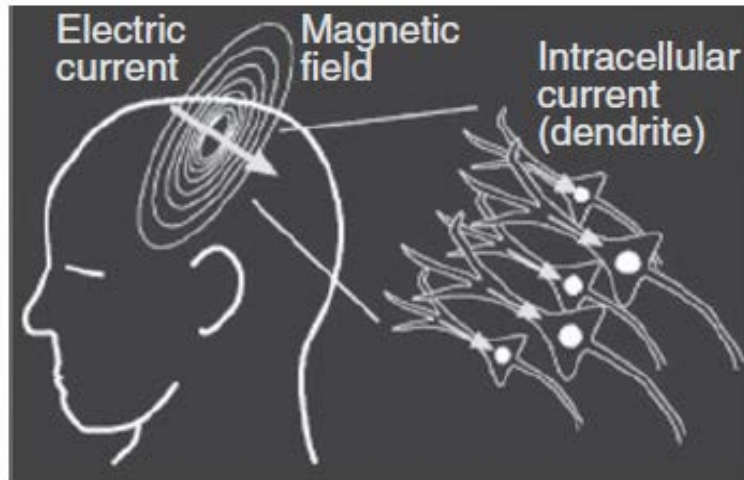
Figure 3.8. **Examples of EEG rhythms and their frequency range**

What's the **characteristic** of EEG?

- EEG signals reflect the **summation** of postsynaptic potentials from many thousands of neurons that are **oriented radially to the scalp**. Currents tangential to the scalp are not detected by EEG.
- Additionally, currents originating deep in the brain are also not detected by EEG because voltage fields fall off with the square of the distance from the source.
- Thus, EEG predominantly captures electrical activity in the **cerebral cortex**, whose columnar arrangement of neurons and proximity to the skull favor recording by EEG.
- The **spatial resolution** of EEG is typically poor (in the square centimeter range) but the **temporal resolution** is good (in the milliseconds range).

Magnetoencephalography (MEG)

- **Magnetoencephalography (MEG)** measures the *magnetic fields* produced by electrical activity in the brain using *superconducting quantum interference devices* (SQUIDs).

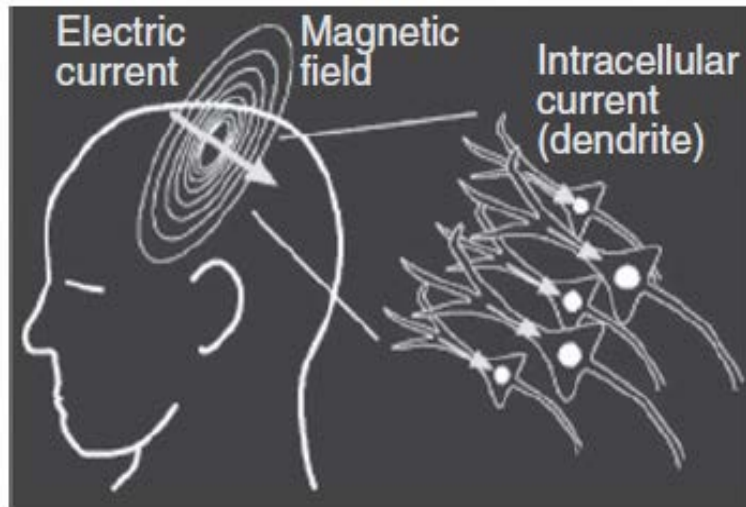


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Both **MEG** and **EEG** signals originate from the ***net effect of ionic currents*** flowing in the dendrites of neurons due to synaptic inputs from other neurons. As shown in Figure 3.9A.



A



B

Figure 3.9. **Magnetoencephalography (MEG).**

(A) Schematic diagram illustrating the orthogonal magnetic field generated by currents in dendrites of similarly oriented cortical neurons.

(B) Example MEG system

Questions

- Does MEG offer high temporal resolution?
 - Yes
- What is the advantage of MEG over EEG?
 - Magnetic fields produced by neural activity are not distorted by the intervening organic matter (such as the skull and the scalp), as is the case with electric fields measured by EEG.
- Does MEG offer better spatial resolution than EEG?
 - Yes.
- What's the disadvantage of MEG, compared to EEG?
 - Expensive, bulky, not portable, require magnetically shielded room

Functional Magnetic Resonance Imaging (fMRI)

Functional Magnetic Resonance Imaging (fMRI)

http://open.163.com/movie/2009/11/1/C/M6I12OHDE_M6I134U1C.htm (5:00-7:00)

- **Principle**

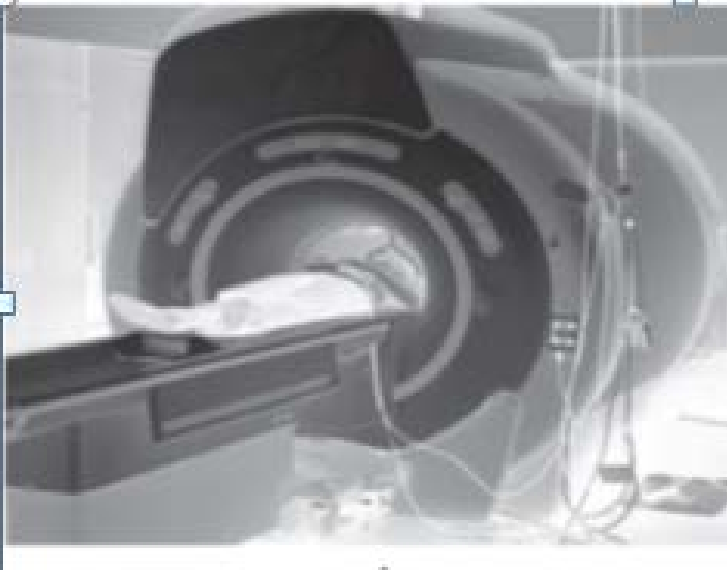
- fMRI **indirectly measures *neural activity*** in the brain by detecting **changes in blood flow** due to increased activation of *neurons* in particular brain areas during specific tasks.
- When neurons become active, they consume more oxygen.
- This hemodynamic response is comparatively slow.
- De-oxygenated hemoglobin is more magnetic than oxygenated hemoglobin.
- fMRI exploits this fact to generate images showing activation in the brain.

- **Recorded signal**

- Given that it measures *oxygenation levels in the blood*, the signal recorded by fMRI is called the **blood oxygenation level dependent (BOLD)** response.

Functional Magnetic Resonance Imaging (fMRI)

Typical settings



Functional Magnetic Resonance Imaging (fMRI)

- **Typical settings**

1. Subjects are made to lie down and their head is positioned inside the fMRI scanner.
2. Subjects may be presented with stimuli:
 - images, sounds or touch
3. Subjects can execute simple actions:
 - pressing a button, moving a joystick, etc.

Functional Magnetic Resonance Imaging (fMRI)

- **Advantage**

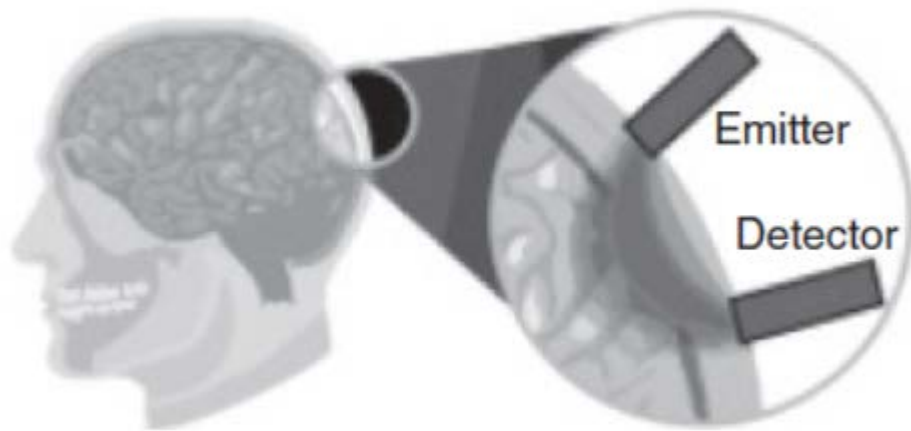
- High spatial resolution
 - 1-3mm

- **Disadvantage**

- Low temporal resolution
 - Several seconds

Functional Near Infrared (fNIR) Imaging

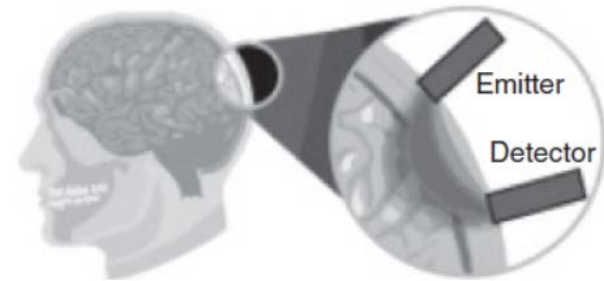
Functional Near Infrared (fNIR) Imaging



Functional Near Infrared (fNIR) Imaging

- **Characteristic**

- Optical technique.
- Measure changes in blood oxygenation level caused by increased neural activity, similar as fMRI.
- Detect near-infrared light absorbance of hemoglobin in the blood with and without oxygen.
- Less cumbersome than fMRI.
- More prone to noise
- Less spatial resolution



Functional Near Infrared (fNIR) Imaging

- **Principle**

- infrared light can penetrate the skull and enter a few centimeters into the cortex.
- Infrared emitters placed on the scalp send infrared light through the skull;
- This light is partly absorbed and partly reflected back through the skull, where it is detected by infrared detectors.
- Infrared light is absorbed differently based on the oxygen content of the blood, providing a measure of the underlying neural activity.

Functional Near Infrared (fNIR) Imaging

- **Principle**

- Similar to EEG, using a number of evenly spaced “optodes” (emitters and detectors) across the head allows one to construct a two-dimensional map of neural activity across the brain surface.



Functional Near Infrared (fNIR) Imaging

- **Properties** (*compared with fMRI or EEG*)
 - Measure neural activity **only close to the skull**, unlike fMRI.
 - **Subjects are not lying down** within an MR scanner, unlike fMRI.
 - **Not susceptible to muscle artifacts**, compared to EEG.
 - **Less expensive** than fMRI.
 - **Portable**, like EEG.

Positron Emission Tomography (PET)

Positron Emission Tomography (PET)

- PET is an older technique for measuring brain activity indirectly by **detecting *metabolic activity***.
- PET measures emissions from radioactively labeled, metabolically active chemicals that have been injected into the bloodstream for transportation to the brain.
- The **spatial resolution** of PET can be comparable to fMRI, but the **temporal resolution** is typically quite low (on the order of several tens of seconds).
- Other **drawbacks** include the need to inject radioactive chemicals into the body and the rapid decay of radioactivity, which limits the amount of time available for experiments.

3.2 Stimulating the Brain

3.2 Stimulating the Brain

- 3.2.1 Invasive Techniques
- 3.2.2 Noninvasive Techniques

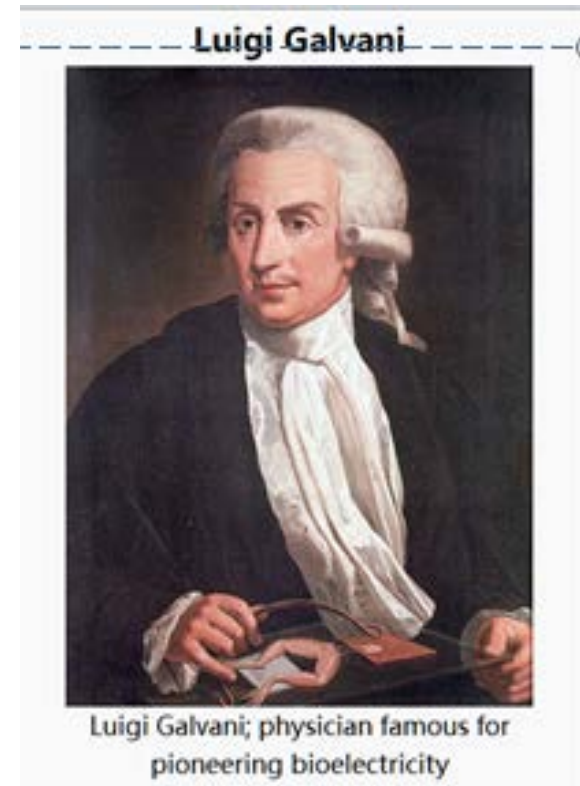
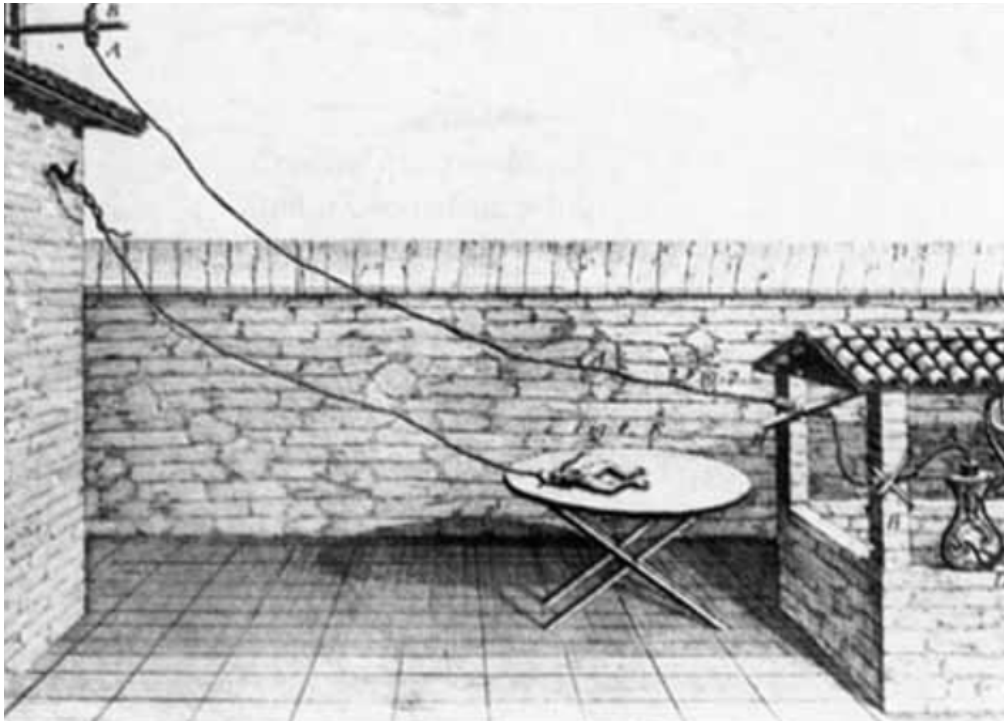
3.2.1 Invasive Techniques

Types of invasive techniques to stimulate the brain

- Microelectrodes
- *Direct Cortical Electrical Stimulation* (DCES)
- Optical Stimulation

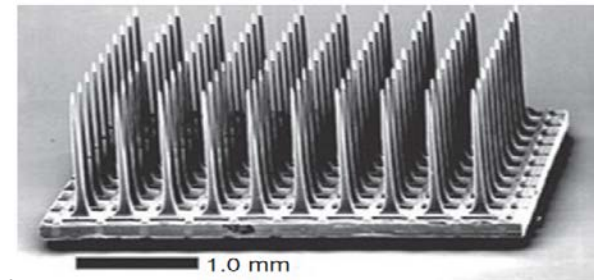
Microelectrodes

- First Experiment:
 - Luigi Galvani, in 1780s



Microelectrodes

- The dominant technology for electrical ***stimulation*** of neurons today uses **the same type** of electrodes used for ***recording*** from neurons, including:
 - **Glass microelectrodes** (intracellularly)
 - **Platinum-iridium microelectrodes** (extracellularly)
 - **Arrays of microelectrodes** (extracellularly)



Direct Cortical Electrical Stimulation (DCES)

- DCES is a semi-invasive method for **stimulating** neurons in the brain, it uses electrodes **on the surface of the cortex** as discussed above for electrocorticography (**ECoG**).
- **Typical settings:**
 - Less than 15 mA
 - Bipolar electrodes.
 - Effect is limited to the several thousands of neurons in the local cortical tissue near the electrode pair.
 - Effects are rapid.

Direct Cortical Electrical Stimulation (DCES)

- **Positive effects:**
 - generating movements
 - causing particular sensations
- **Negative effects:**
 - disruption of a movement or behavior
- **Applications:**
 - typically used in **a clinical setting** for *mapping the location* of *sensory, motor, memory, and language* functions in the brains of neurosurgery patients.
 - Its potential for **providing direct feedback** during brain-computer interfacing remains to be explored.

Optical Stimulation

- **Background**

- Laser illumination can produce excitation in a neuron. (Fork, 1971)

- **Approaches**

- **two-photon laser illumination**

- offers a method to selectively excite single neurons

- **optogenetic stimulation**

- could provide a means to selectively excite only a specific class or classes of neurons that have been genetically altered using cell-specific methods.

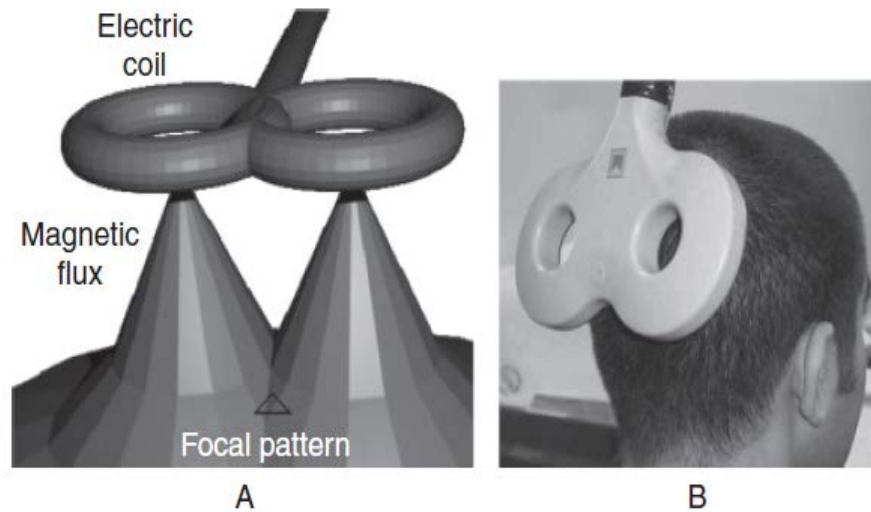
3.2.2 Noninvasive Techniques

3.2.2 Noninvasive Techniques

1. Transcranial Magnetic Stimulation (TMS)
2. Transcranial Ultrasound

Transcranial Magnetic Stimulation (TMS)

Transcranial Magnetic Stimulation (TMS)



- Figure 3.11. **Transcranial magnetic stimulation (TMS).**
- (A) Schematic illustration of electrical stimulation produced by electromagnetic induction using a “butterfly” coil .
- (B) TMS of visual cortex of a subject using a butterfly coil.

Transcranial Magnetic Stimulation (TMS)

Principle

- TMS relies on the close relationship between electricity and magnetism and the process of **electromagnetic induction**:
 - when current is passed through a coil of wire, a magnetic field is generated perpendicular to the current flow in the coil. If a second coil is placed within the magnetic field, a current is generated in a direction opposite the first flow.
- TMS exploits this phenomenon by placing a plastic-enclosed coil of wire next to the skull to produce a rapidly changing magnetic field oriented orthogonal to the plane of the coil.
- The magnetic field passes unimpeded through the skin and skull and, by the principle of electromagnetic induction, produces an electric current in the brain that activates populations of neurons.

Transcranial Magnetic Stimulation (TMS)

Suitable field

- The magnetic field produced by TMS is believed to penetrate to a maximum depth of about 3 to 5 centimeters into the brain, in the area directly adjacent to the coil.
- The technique is therefore suitable only for activating neurons in the superficial layers of the brain.

Transcranial Magnetic Stimulation (TMS)

Advantage & Disadvantage

- **Advantage:**

- A major advantage of TMS is that it is noninvasive and its use is not restricted to patients.

- **Disadvantage:**

- Its disadvantages include the relatively high power requirements and poor localization of the area of stimulation compared to invasive techniques such as microelectrodes and DCES.

Transcranial Ultrasound

What's the ultrasound?

- Ultrasound is a mechanical pressure wave (sound wave) having a frequency above the range of human hearing (>20 kHz).

What's the unique property of the ultrasound?

- Ultrasound has the favorable property that it can be transmitted through solid structures, including bone and soft tissues, making it well suited for noninvasive medical applications.
- It is known that high-intensity ultrasound ($> 1\text{W}/\text{cm}^2$) affects neural activity through thermal effects, but such stimulation can cause tissue damage.
- Fortunately, researchers have found that low-intensity ($< 500\text{ mW}/\text{cm}^2$) pulsed ultrasound can also influence neural activation but without thermal effects or tissue damage.

Principle of Ultrasound Technology

- The exact mechanisms underlying the effects of ultrasound on neural activation are unknown.
- But it has been suggested that ultrasound may affect neural ion channels with mechanically sensitive gating kinetics or produce fluid-mechanical effects on the cellular environments of neurons, thereby affecting their resting membrane potentials.

Advantage & Disadvantage

- **Advantage**

- Pulsed ultrasound may offer an advantage over TMS in terms of spatial resolution in that it can stimulate brain regions 1–2 mm in diameter, compared to 1 cm or greater in the case of TMS.

- **Disadvantage**

- It remains to be seen whether it can be used as part of a noninvasive BCI system for delivering targeted feedback to specific brain areas in closed-loop BCI tasks.

3.3 Simultaneous Recording and Stimulation

Although most current BCIs **only record** ongoing neural activity to control devices and provide visual or tactile feedback, some researchers are exploring the possibility of **simultaneously** recording neural signals and providing direct feedback through neural stimulation.

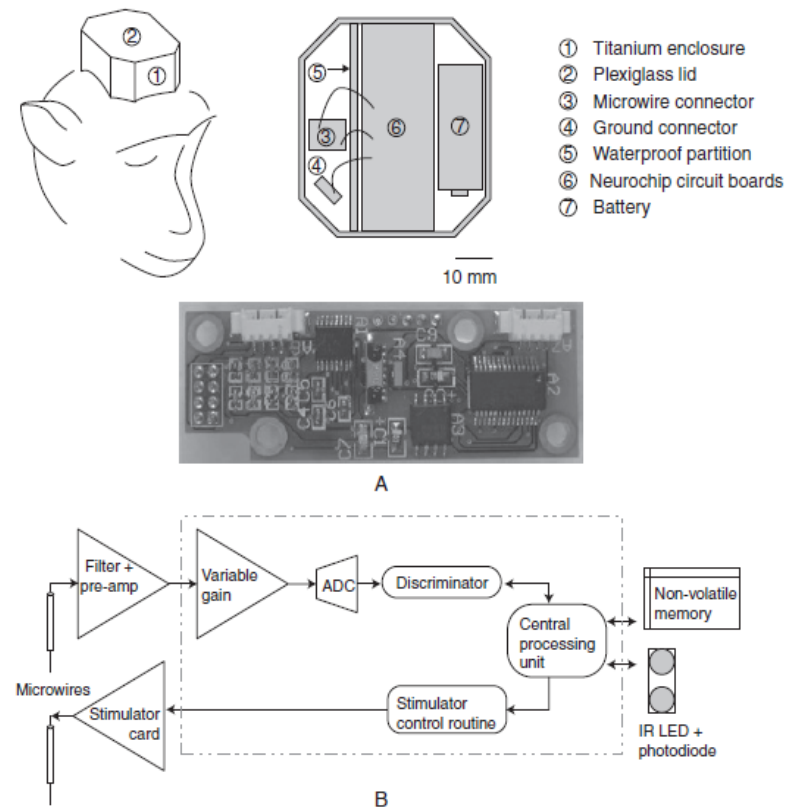
Two possible approaches being explored include: **Multielectrode Arrays** and more sophisticated implantable chips, such as the **Neurochip**.

3.3.1 Multielectrode Arrays

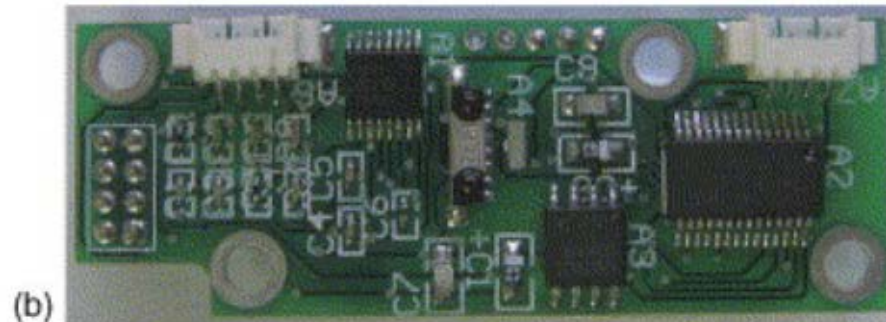
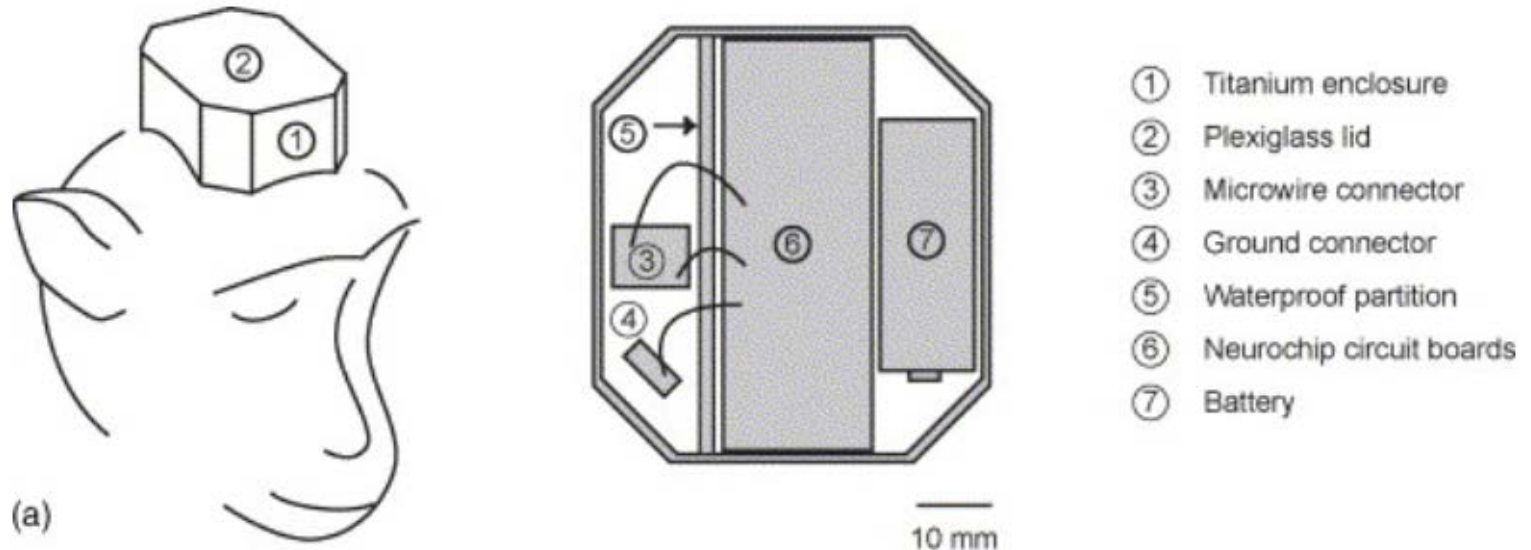
- As described above, microelectrodes used for recording spiking activity can also be used to deliver depolarizing or hyperpolarizing current to excite or inhibit neurons.
- Thus, in a multielectrode array, some electrodes can be set aside for recording and others may be used for stimulation.
- We will explore such a use of multielectrode arrays in Chapter 11 .

3.3.2 Neurochip

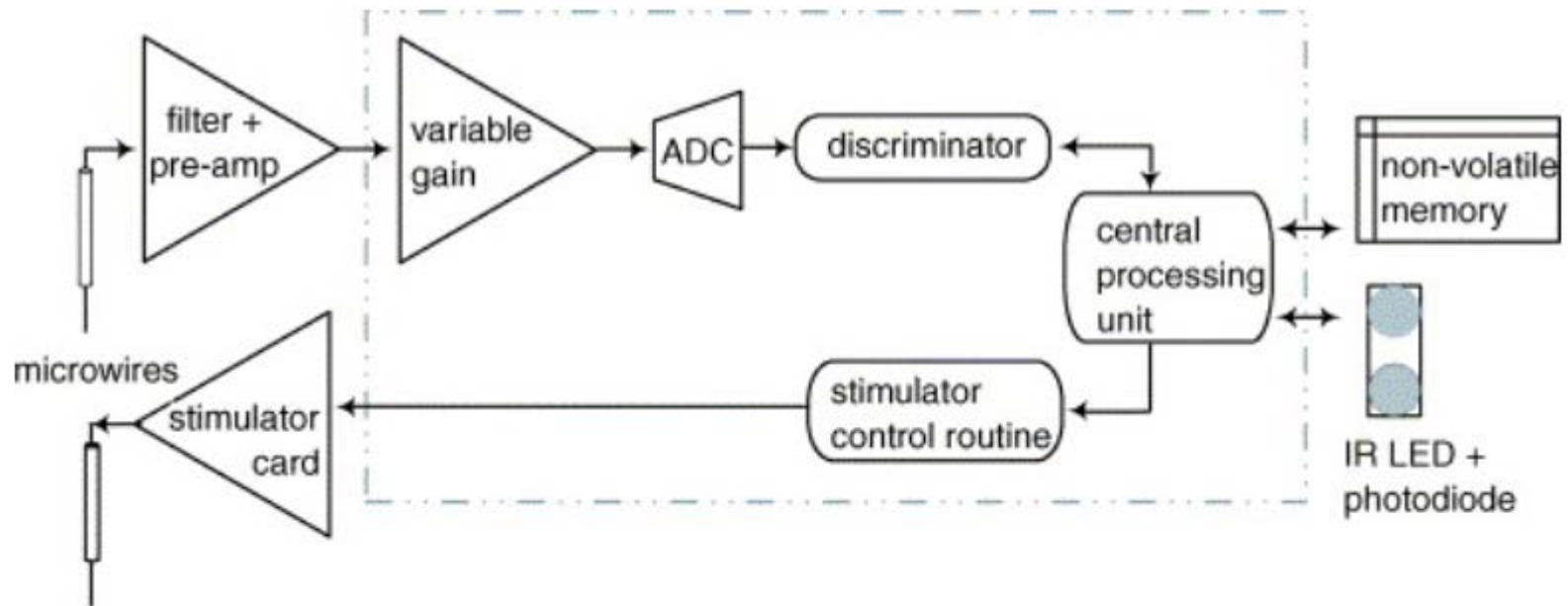
- Figure 3.12. **Neurochip for simultaneous recording and stimulation of neurons.**
- (A) Components of the implant containing Neurochip.
- (B). Architecture of the Neurochip , showing analog and digital components, on-chip memory, and IR LED and photodiode for wireless communication of up to 1 meter.



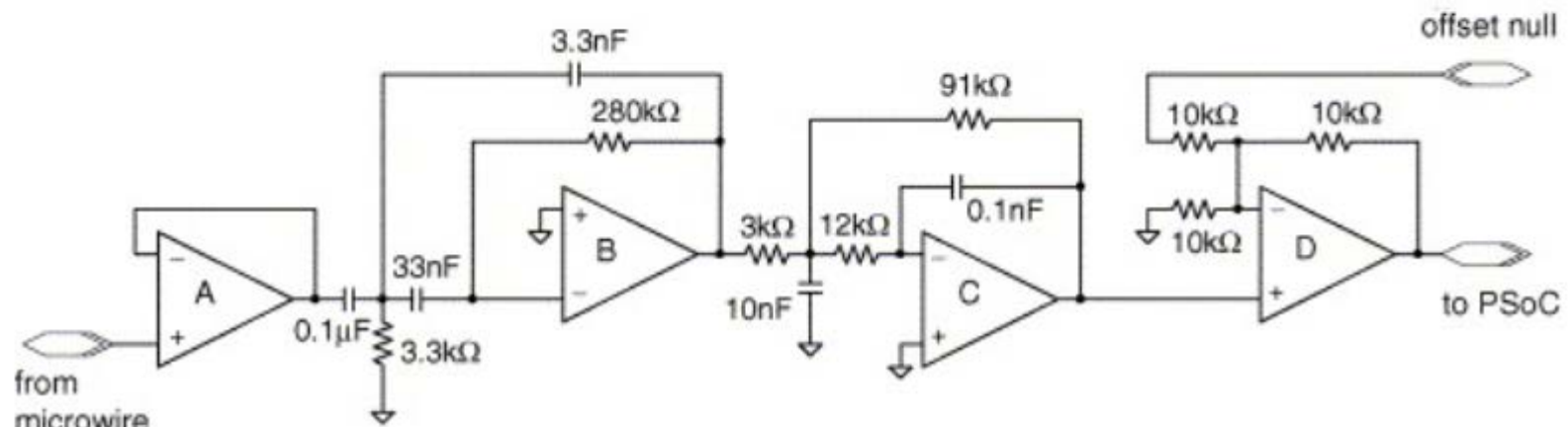
3.3.2 Neurochip



J. Mavoori, A. Jackson, C. Diorio, and E. Fetz, "**An autonomous implantable computer for neural recording and stimulation in unrestrained primates,**" *Journal of Neuroscience Methods*, vol. 148, pp. 71-77, 2005.



(a)



(b)

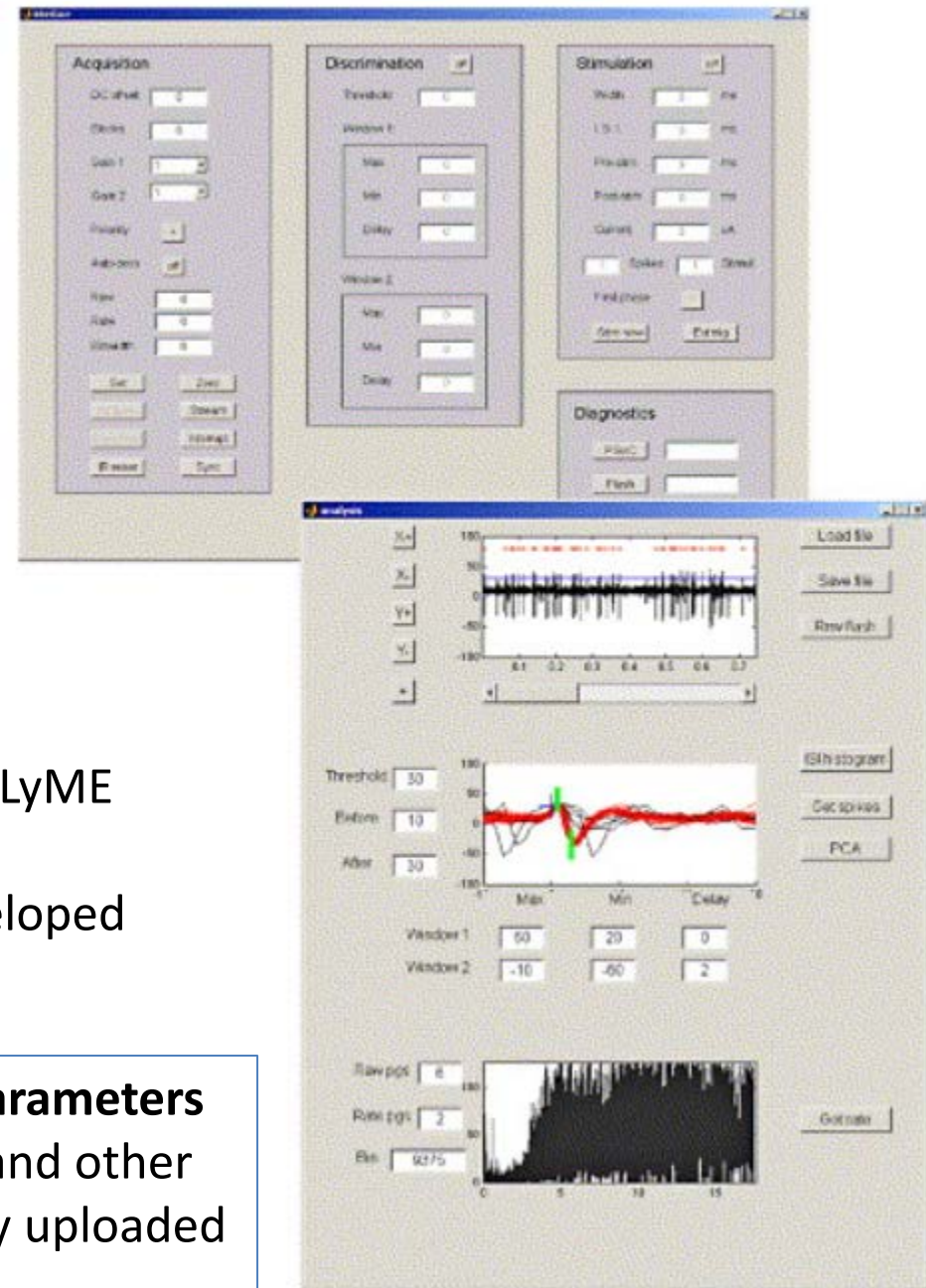


Fig. 3. User interfaces to neurochip.
 (a) a palm user interface developed in LyME (Calerga Sarl)
 (b) a similar interface for a PC developed in Matlab (Mathworks Inc.).

The interfaces facilitate **transmission of parameters** for recording, discrimination, stimulation and other computations to the neurochip and display uploaded data.

3.3.2 Neurochip

- The Neurochip has been used in monkeys to demonstrate:
 - consistent activation of a group of neurons in correlation with the activation of another can cause a strengthening of connections between the two groups of neurons.
- We will examine the use of the Neurochip for BCI applications in Chapter 11.
- video http://open.163.com/movie/2009/11/A/1/M6I12OHDE_M6I3L6TA1.html

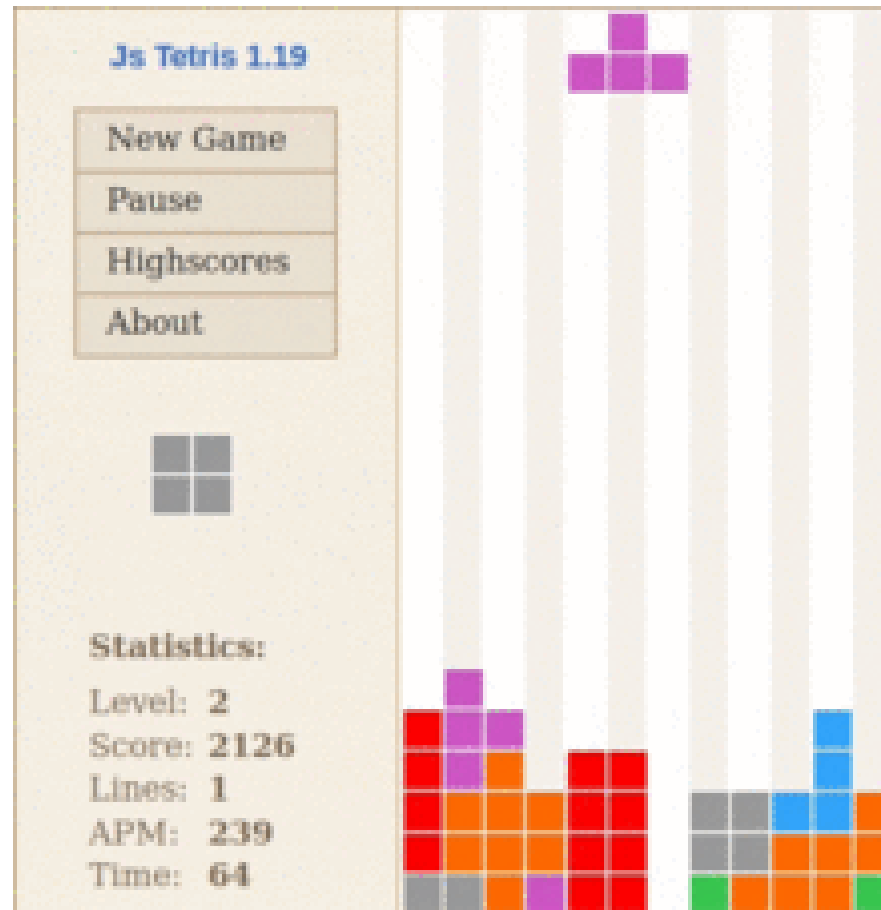
Recent Research by Rao's Group

- L. Jiang, A. Stocco, D. M. Losey, J. A. Abernethy, C. S. Prat, and R. P. N. Rao, "**BrainNet: A Multi-Person Brain-to-Brain Interface for Direct Collaboration Between Brains,**" 2018.



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Recent Research by Rao's Group



Recent Research by Rao's Group

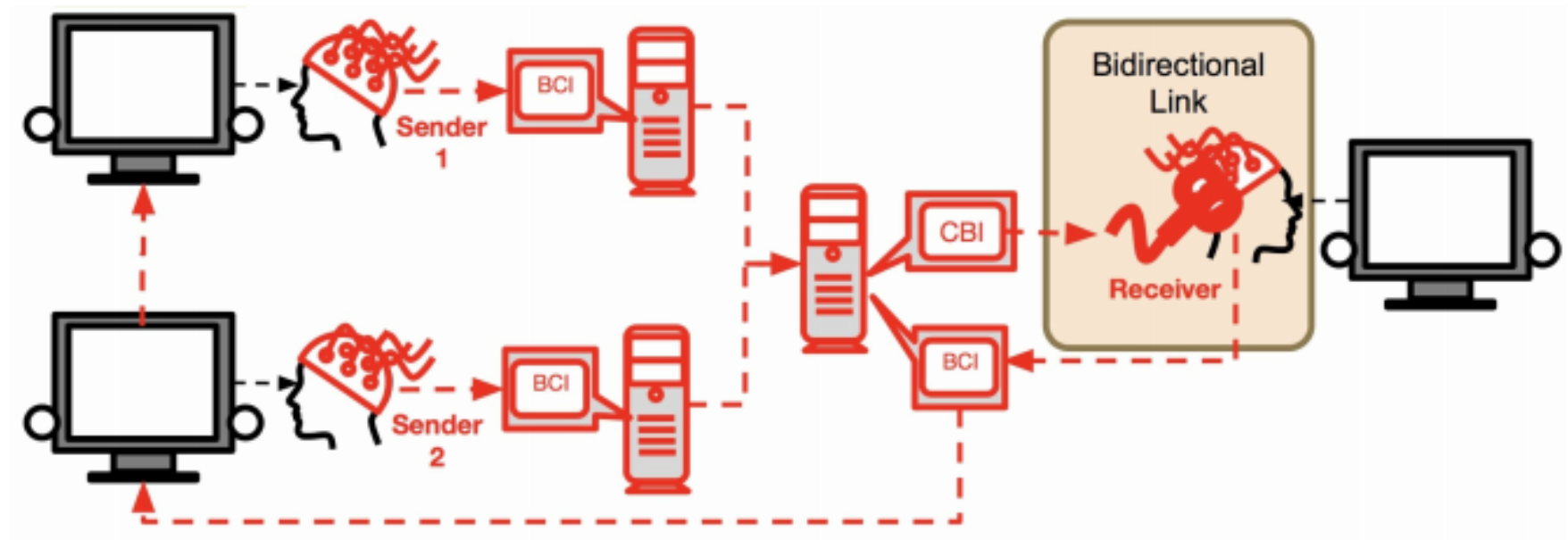


Figure 1. Architecture of BrainNet. Two participants ("Sender 1" and "Sender 2") each use a Brain-Computer Interface (BCI) based on EEG to convey information about a collaborative task (here, a Tetris-like game) directly to the brain of the third participant ("Receiver"). Information from each Sender is transmitted over the internet to the Receiver's brain via a Computer-Brain Interface (CBI) based on TMS. After consciously processing the two inputs from the Senders, the Receiver uses a BCI based on EEG to execute an action in the task. The Senders see the result of this action on the task and have another opportunity to convey to the Receiver's brain new information to potentially rectify an incorrect choice in the first round. While our experiment only used two rounds, BrainNet allows an arbitrary number of interactions between the Senders and the Receiver as they collaborate to solve a task.

Recent Research by Rao's Groups

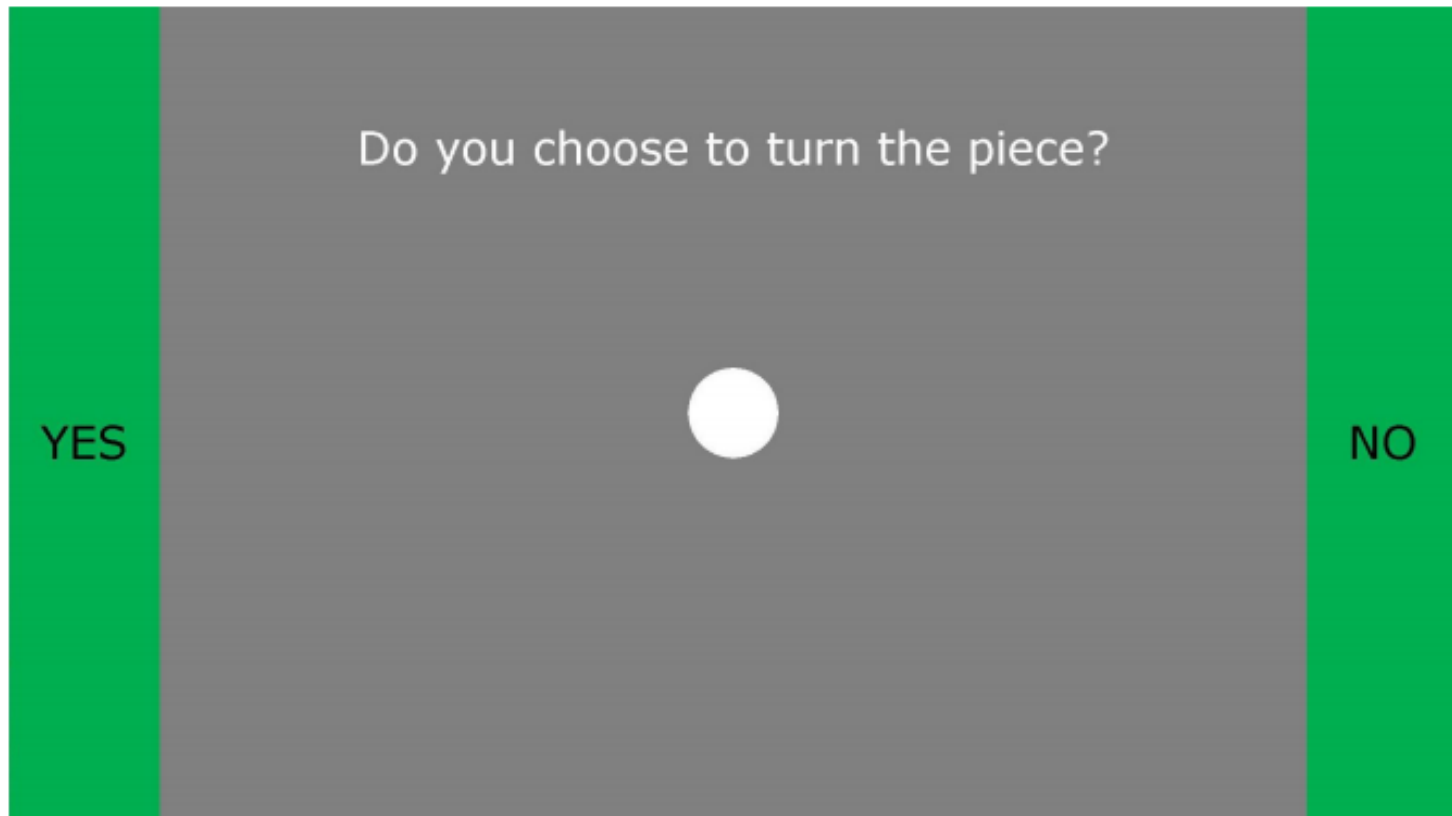


Figure 8. SSVEP-Based EEG Brain-Computer Interface used by the Senders and the Receiver. Participants conveyed their decisions regarding whether or not to rotate the current piece by controlling a cursor (white filled circle) using steady state visually evoked potentials (SSVEPs) based on EEG. Participants focused on a flashing LED to the left of the screen (depicted as a circle attached to the screen in Figure 1) to move the cursor leftwards towards the “Yes” side. Focusing on the LED to the right of the screen (flashing at a different frequency) caused the cursor to move rightwards towards the “No” side. If the cursor reached the green “Yes” bar, the interface interpreted the participant’s decision to be rotation of the piece (by 180 degrees). If the cursor reached the “No” bar, the interface took the participant’s decision to be to keep the piece’s current orientation.

Recent Research by Rao's Group



Minghui Shi, Xiamen University, China

3.4 Summary

3.4 Summary

- **Invasive methods**
 - Single electrode (**Recording and Stimulating**)
 - Multielectrode Array (**Recording and Stimulating**)
 - Neurochip (**Recording and Stimulating**)
 - DBS, DCES, Optical Stimulation (**Stimulating**)
- **Semi-invasive techniques**
 - ECoG (**Recording and Stimulating**)
- **Noninvasive techniques**
 - EEG, MEG, fMRI, fNIR, PET (**Recording**)
 - TMS, Transcranial Ultrasound (**Stimulating**)

Thank you for your attention!